

Intelligent Table Tennis Analytics for Enhanced Performance Tracking

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Abstract

This project presents the development of a system for table tennis player performance analytics, which utilizes foundation models, computer vision algorithms, and audio processing techniques. The system processes video data acquired from world table tennis match footage. An initial auto-labeling pipeline, employing CLIP-based semantic scoring and frame selection, is used to identify salient game segments for analysis. Methodologies explored include the application of foundation models for scene understanding tasks such as table detection with subsequent geometric processing, and for contextual player identification. Computer vision modules were developed for: (1) player tracking, using custom-trained object detectors and appearance-based re-identification; (2) ball segmentation and tracking, employing a spatio-temporal 3D U-Net with motion filtering based on player presence. Additionally, (3) audio analysis techniques are applied for classifying acoustic events like racket and table hits from Mel-Frequency Cepstral Coefficient (MFCC) features using convolutional neural networks; and (4) player movement patterns are visualized through spatial heatmaps. This integrated approach is intended to yield multi-faceted data, with the goal of offering insights into player performance and game dynamics in table tennis.

Code is available at: <https://github.com/PratyushCS12/Table-Tennis-2025-SMAI-SP-25.git>

Motivation

The motivation for this project is to contribute to the automation and enhancement of table tennis analysis. Table tennis is a fast-paced sport where manual annotation can be time-consuming and subjective. Automated systems may provide objective data on player behavior, ball movement, and game events, which can be useful for coaches in refining strategies, for players in improving skills by understanding patterns, and for broadcasters in creating content. The development of such a system also allows for exploration of AI applications in addressing challenges common in sports analytics, such as noisy environments, rapid motion, and the need for multi-object tracking and event recognition.

Literature Review

The methodologies explored in this project draw from established and developing research areas within machine learning, computer vision, and signal processing. The use of **foundation models** is being explored; models like CLIP (Contrastive Language-Image Pre-Training) are utilized for their zero-shot or few-shot capabilities in semantic understanding, which can assist in tasks like automated frame selection. For **object detection**, including players and potentially the table, large vision models pre-trained on extensive datasets offer feature extraction and detection capabilities. Custom-trained models, such as **You Only Look Once (YOLO)** and its variants, are a common choice for object detection when domain-specific annotated data is available. For **fine-grained segmentation** of small, fast-moving objects like the ping pong ball, **U-Net architectures**, particularly **3D U-Nets** that process spatio-temporal information, are often used. Player **tracking-by-detection** paradigms are common, often combining object detectors with appearance features (e.g., color histograms in HSV space) and motion models for identity maintenance. **Audio event detection** frequently involves extracting time-frequency representations like **MFCCs**, which are then classified using machine learning models such as traditional SVMs or **Convolutional Neural Networks (CNNs)**, capable of learning hierarchical patterns from spectrograms. Finally, player movement analysis through **spatial heatmaps** is a visualization technique used in sports analytics. Efficient data handling, including preprocessing strategies like video chunking and frame caching, is important for managing the large volumes of data inherent in video analysis.

Problem Statement

The objective of this project is to develop an automated system for analyzing table tennis videos, aiming to identify and characterize key elements and events within the game. This is addressed by deconstructing the problem into several interconnected subtasks:

1. **Automated Dataset Curation and Scene Establishment:** To develop a pipeline for processing raw table tennis match footage, attempting to automatically select relevant frames or segments using semantic understanding (e.g., via CLIP scoring), and to establish foundational scene geometry, including the detection of the playing table, potentially using foundation models.

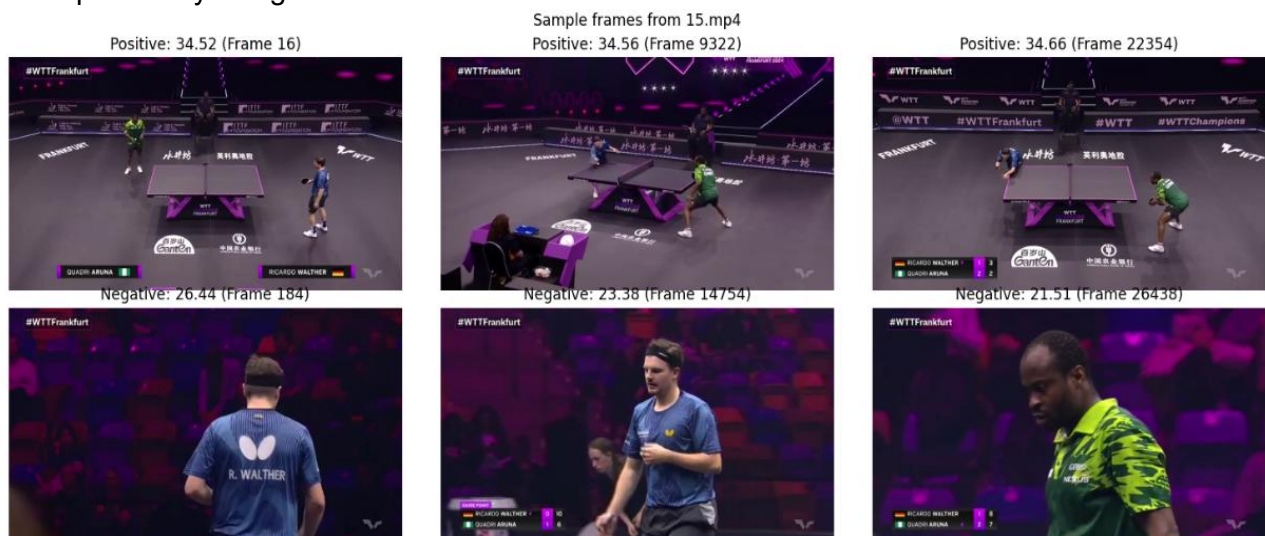


Figure: Positive frames are the region where the gameplay action happens

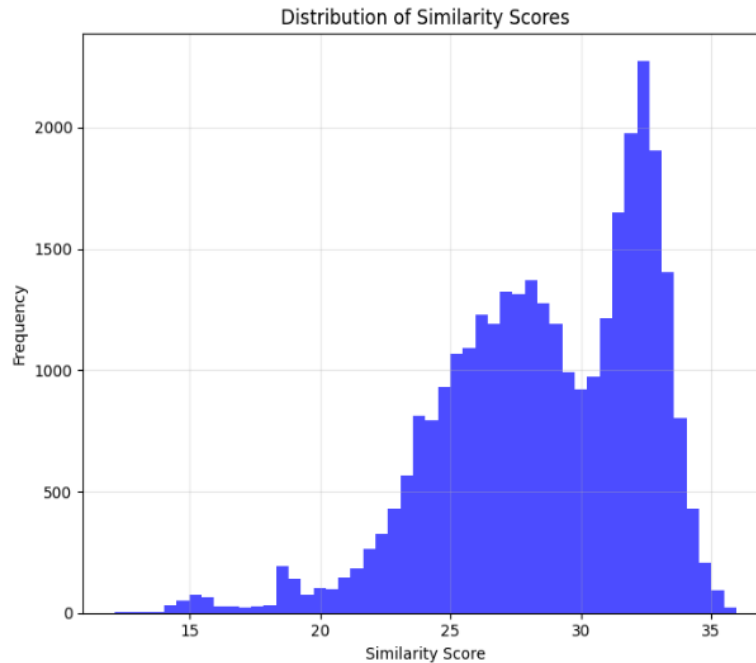
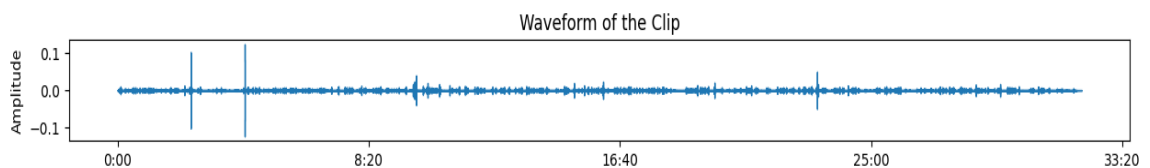


Figure : Distribution of similarity with respect to the prompt *“Players playing table tennis on a table tennis board. No close face zoom audience focus on players playing all the players are visible in a frame. Not a single person serving”*

2. **Audio Event Classification:** To attempt to automatically identify and classify distinct sound events relevant to understanding game flow, specifically "racket hits," "table hits," and periods of "no hit," by processing the audio stream.



3. **Player Detection and Tracking:** To detect table tennis players in video frames, assign unique identities, and track their movements during a match, addressing challenges like occlusions or rapid motion.

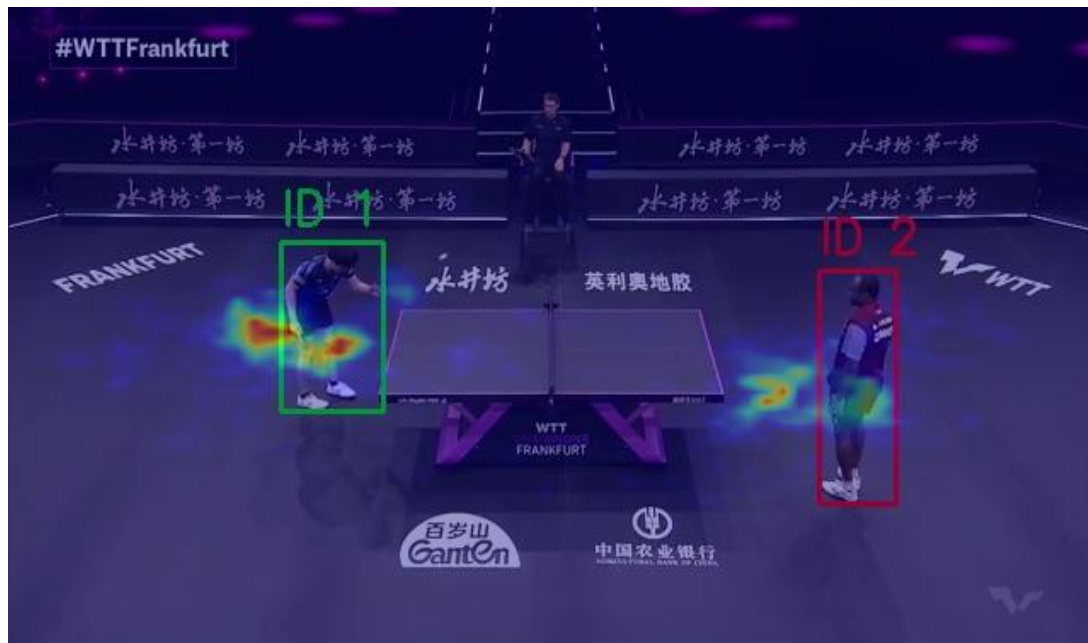


Figure: Player Detection and Tracking Heatmap

4. **Ball Tracking and Segmentation:** To detect and segment the ping pong ball in video sequences, aiming to distinguish it from visual clutter and player movements, with the goal of enabling trajectory reconstruction.

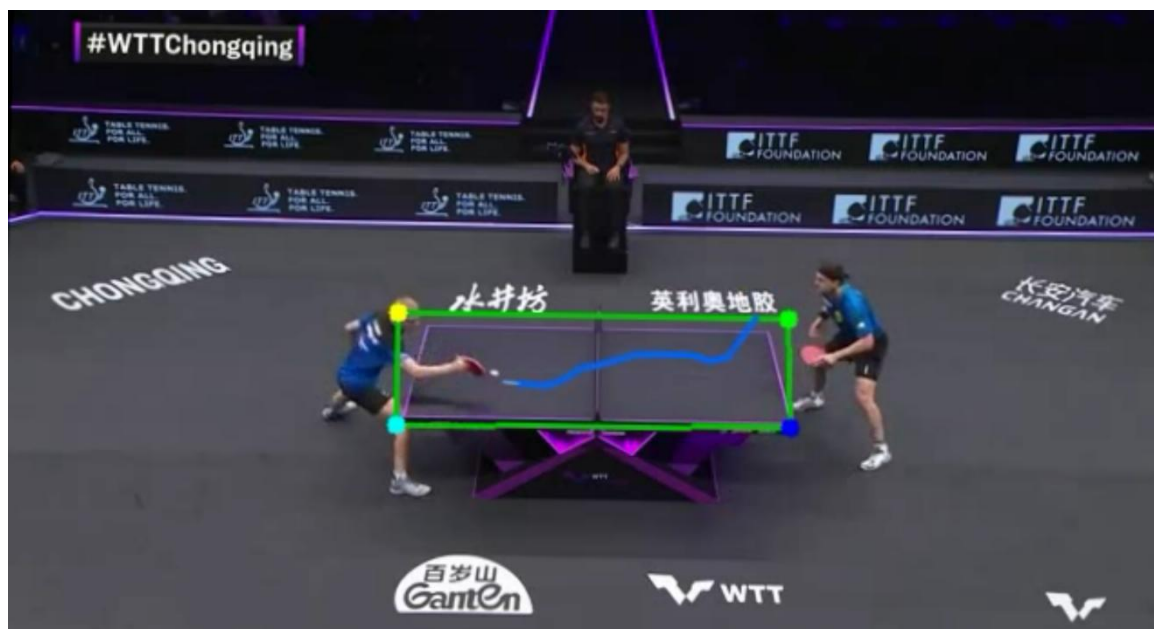


Figure: The above figure show **ball trajectory** along with table key points



Figure: Table Boundary data generation with foundation models and image processing (used for generating keypoint dataset)

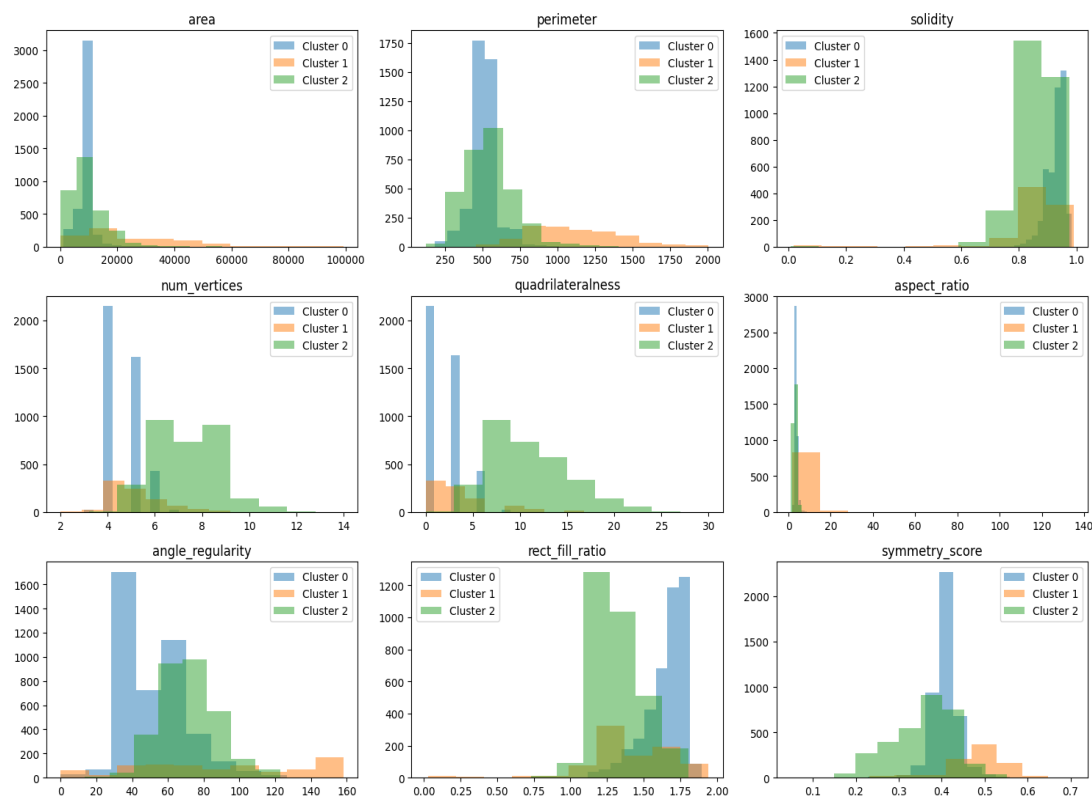


Figure: Making use of various geometric metric for separating the actual table contours detected by foundation model from the false detection

Methodology

Dataset Acquisition and Preprocessing The system utilizes video footage sourced from publicly available world table tennis matches. An initial step involves an automated data curation pipeline. This pipeline employs a foundation model, CLIP, to perform semantic frame selection. By scoring frames against textual prompts relevant to active gameplay, the system attempts to identify and prioritize frames that are likely to contain game events, aiming to streamline subsequent analysis.

The primary dataset thus consists of these selected video segments (e.g., MP4 format) and associated metadata. Further annotations, such as "positive frames" for ball presence or player actions, are either derived from this auto-labeling process or provided separately in JSON format. Audio is extracted from video files (commonly as WAV) for independent analysis. The system generates intermediate data products, including extracted image frames (JPEG/PNG), cached frames for model inputs, and structured outputs like CSV files for tracking data and event logs.

Subtasks and Implementation Details

1. Scene Establishment: Table Detection and Canonicalization:

- An important preliminary step for contextual understanding is the detection and geometric analysis of the table tennis table. A foundation model suitable for object detection or visual grounding is considered for this task.
- Upon detecting the table, further processing may be applied to its bounding box or segmentation mask. This could involve perspective correction or identifying key points, aiming to establish a reference frame for spatial analysis.

2. Frame Extraction and Preparation for Downstream Models:

- Following initial frame selection and table detection, a frame extraction module processes relevant video segments. To manage computational resources, video processing can be performed in temporal chunks.
- Extracted frames are resized to resolutions suitable for specific models and stored as image files.
- This frame data is then partitioned into training and validation sets for various supervised learning tasks, with attention to stratified sampling where appropriate.

3. Player Detection and Bounding Box Generation:

- For identifying players, a dual approach is explored:
 - A zero-shot object detection model (e.g., a Grounding DINO variant), acting as a foundation model, uses textual prompts for initial or contextual player detection.
 - A custom object detector (e.g., a YOLO variant) is fine-tuned on a dataset of player images and bounding boxes.
- The system processes frames in batches, with detections filtered by confidence scores. Bounding box information is stored for tracking and analysis.



4. Player Tracking:

- Player tracking builds upon detections from the custom-trained YOLO model, focusing on two-player scenarios.
- An appearance-based re-identification strategy is employed. HSV color profiles are extracted from player bounding boxes to serve as appearance signatures.
- In subsequent frames, detected player instances are matched to these reference signatures using an HSV color distance metric, aiming to maintain consistent player IDs.
- Here Outlier and false detection is removed by threshold on frame wise maximum movements.

5. Ball Tracking and Segmentation:

- A 3D U-Net architecture is implemented for segmenting the table tennis ball, using spatio-temporal information from sequences of frames (typically 3 consecutive frames).
- The motion data was created using image processing operations by frame differencing, followed by sequence of image processing operation like otsu thresholding, dilation, rejecting large areas based on size of component.
- We observe that 3D unet training generates better motion segmentation results as compared to the case where the camera is in motion because in those cases the image processing operation fails but our network generalizes in unstable frames.
- **Data Handling:** Relevant frames for sequence generation are pre-cached.
- **Contextual Filtering:** To reduce false positives from player movements, a pre-trained object detector identifies persons near potential ball motion. Candidate ball regions detected through motion analysis (frame differencing, thresholding,

connected components) that overlap significantly with detected persons are filtered out.

- **Training Data Generation:** For valid motion candidates, segmentation masks are generated. These image sequences and masks are resized (e.g., to 128x128) and stored as numerical arrays for training the 3D U-Net.
- **Model Training:** The 3D U-Net is trained to predict segmentation masks from input sequences, with parameters like batch size, learning rate, and early stopping configured.
- The **3D Unet** segments out various candidates for motion segmentation using these candidates we compute trajectories and only consider those candidates that overlap with the table tennis table surface region.

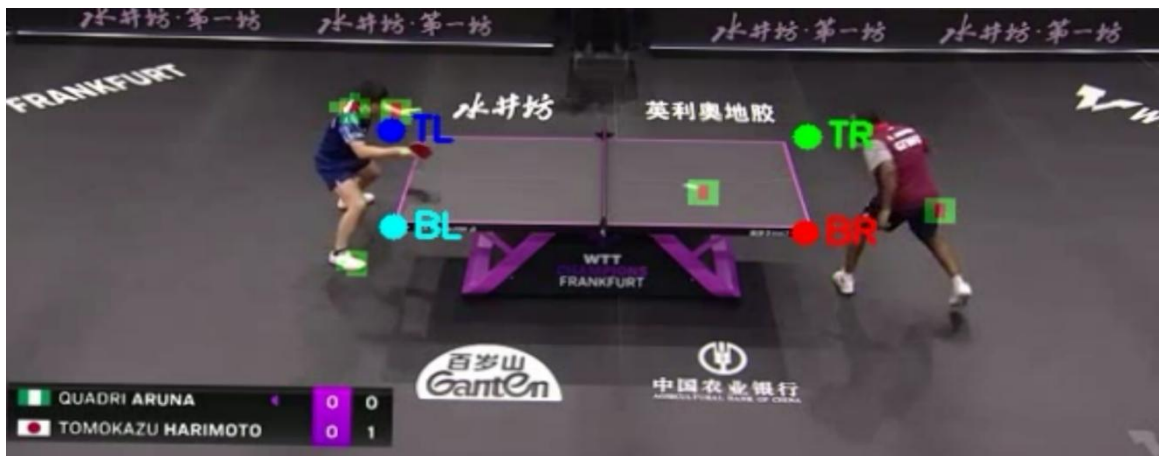


Figure: The above figure shows the output of the 3D UNet for segmenting out the motion and the ball motion is also segmented out along with the table keypoint. These information are used for segmenting out the ball trajectory.

Figure : The above figure shows the output of the image processing algorithm used for the preparation of noisy labels. This approach produces data used for training the 3D unet. The 3D Unet generalizes in unstable video with camera motion.

6. Audio Event Classification:

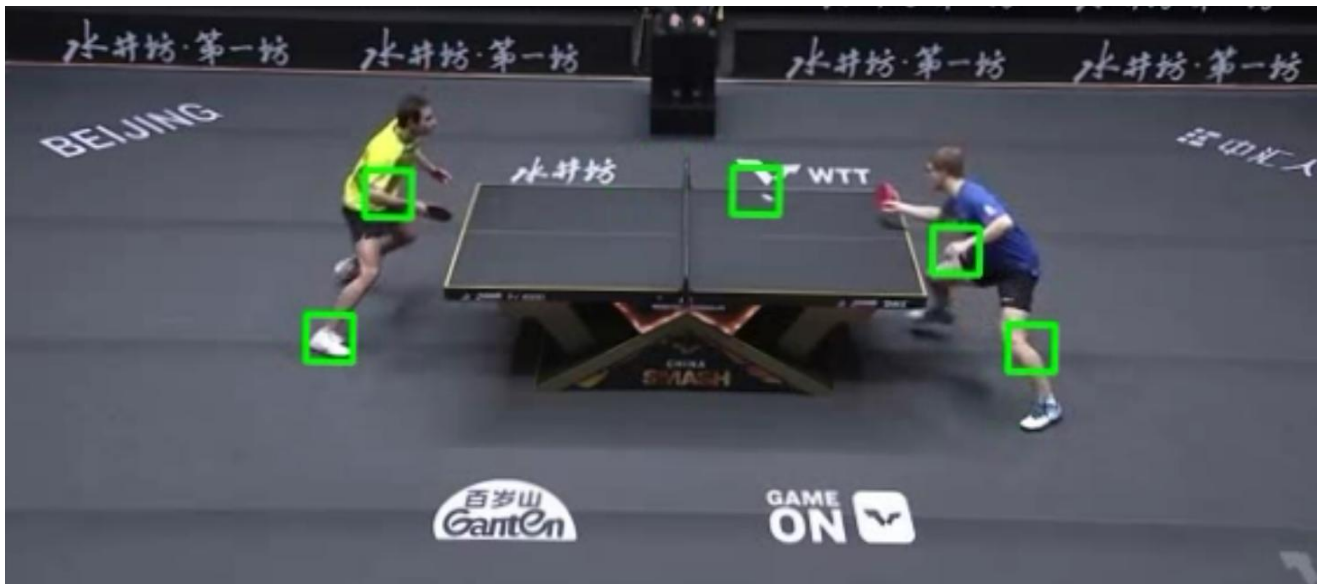
- Audio streams are processed to detect and classify acoustic events. The recordings are first converted to mono and resampled to 22.05 kHz. Applied spectral flux onset detection to identify candidate hit times.
- Short audio clips (e.g., 0.1 s) around these onsets are extracted. A heuristic based on spectral centroid is used to differentiate 'racket' from 'table' hits. 'No-hit' samples are taken from quiet segments.

- A 100-ms window (20 ms before, 80 ms after) is extracted as a hit clip for each onset. “No-hit” clips of the same length are sampled from segments at least 50 ms away from any onset and filtered by RMS (< 0.01) and peak amplitude (< 0.005).
- Each clip is converted into a normalized MFCC spectrogram (32 coefficients $\times$ fixed time frames), producing a 3D tensor ($32 \times T \times 1$) for CNN input.
- A compact 2D CNN: Conv2D $\rightarrow$ MaxPool $\rightarrow$ Conv2D $\rightarrow$ MaxPool $\rightarrow$ Flatten $\rightarrow$ Dense uses sparse categorical crossentropy to classify clips into “racket hit,” “table hit,” or “no hit.”

Experiment

1. Dataset Curation and Scene Setup:

- The CLIP-based auto-labeling pipeline is applied to table tennis match videos to identify frames depicting active gameplay.
- A foundation model is used on these frames to attempt table detection. The outcome of table detection (e.g., qualitative assessment of alignment) is noted.



2. Audio Analysis:

- The audio processing experiment classifies detected onsets.
- MFCCs are extracted from 0.1-second audio clips.
- A 2D CNN is trained (e.g., 20 epochs, batch size 32). Outputs include the trained model, a log of audio events, and audio clips.
- Performance visualization includes training/validation accuracy/loss curves, a confusion matrix, and PCA of features.

3. Player Tracking:

- **Model Training:** A YOLO model is fine-tuned for “table tennis player” detection. Training parameters include epochs (e.g., 10), image resolution (e.g., 640x640), and batch size (e.g., 32).
- **Tracking Execution:** The trained YOLO model, with HSV color-based re-identification, tracks players in videos.

- **Movement Visualization:** Player tracking data is used to generate 2D heatmaps illustrating spatial occupancy. Total distance covered by players is also calculated.
4. **Video Segmentation (Ball) & Contextual Understanding:**
- **Data Preparation:** The frame extraction module prepares a general dataset. Player detection provides context for ball segmentation data generation.
 - **Ball Segmentation Data Generation:** Sequences of frames around positive ball instances are created. Motion analysis proposes ball candidates, filtered using player location data. Masks are generated.
5. **Ball Tracking (Inference):**
- It takes around 30 seconds to process 60 seconds of the match (can vary from gpu platform to platform)

```

--- Processing video: /content/drive/MyDrive/SMAI_Project_dataset/RightVideo/1.mp4 ---
Video FPS: 25.0, Dimensions: 640x360
Processing up to 1500 frames (approx. 60 seconds).
Inferencing 1.mp4: 100%|██████████| 1500/1500 [00:30<00:00, 49.99it/s]
Finished processing /content/drive/MyDrive/SMAI_Project_dataset/RightVideo/1.mp4.
Processed and wrote 1498 frames to video.
Output video saved to: /content/drive/MyDrive/SMAI_Project_dataset/1_extrapolated_trajectory.mp4

--- Processing video: /content/drive/MyDrive/SMAI_Project_dataset/RightVideo/2.mp4 ---
Video FPS: 25.0, Dimensions: 640x360
Processing up to 1500 frames (approx. 60 seconds).
Inferencing 2.mp4: 100%|██████████| 1500/1500 [00:28<00:00, 52.38it/s]
Finished processing /content/drive/MyDrive/SMAI_Project_dataset/RightVideo/2.mp4.
Processed and wrote 1498 frames to video.
Output video saved to: /content/drive/MyDrive/SMAI_Project_dataset/2_extrapolated_trajectory.mp4

--- Processing video: /content/drive/MyDrive/SMAI_Project_dataset/RightVideo/3.mp4 ---
Video FPS: 25.0, Dimensions: 640x360
Processing up to 1500 frames (approx. 60 seconds).
Inferencing 3.mp4: 100%|██████████| 1500/1500 [00:29<00:00, 50.89it/s]
Finished processing /content/drive/MyDrive/SMAI_Project_dataset/RightVideo/3.mp4.
Processed and wrote 1498 frames to video.
Output video saved to: /content/drive/MyDrive/SMAI_Project_dataset/3_extrapolated_trajectory.mp4

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Result

The system's performance is assessed through a combination of quantitative metrics and qualitative visualizations for each sub-task, where feasible.

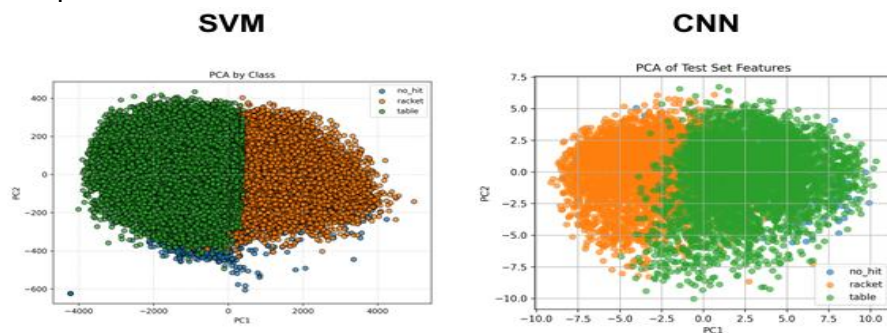
Tables (Illustrative)

- **CLIP-based Frame Selection Efficacy:** A table indicating precision/recall of the auto-labeling pipeline in identifying relevant game segments, if compared to a manually annotated ground truth subset.
- **Foundation Model Table Detection Outcome:** A qualitative description or, if ground truth is available, metrics like Intersection over Union (IoU) for table detections.

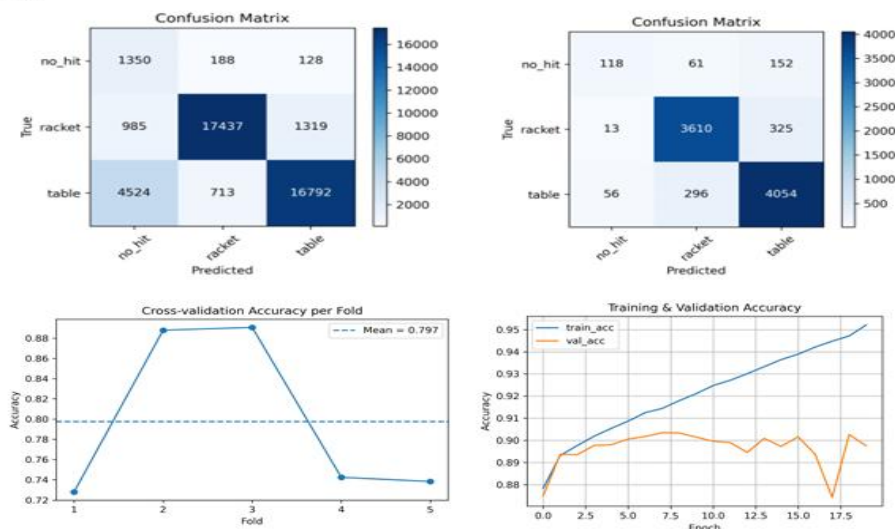
- **Audio Event Classification Performance:** A classification report (precision, recall, F1-score per class) and overall accuracy for audio event detection on a test set.
- **Ball Segmentation Performance (3D U-Net):** Segmentation quality metrics such as Dice Score or IoU for ball segmentation, evaluated against a test set of annotated sequences.
- **Player Tracking Statistics:** Summary of metrics such as average track length or total distance covered per player.

Graphs and Plots and Visualization

1. **Dataset and Scene Setup:**
 - Visual examples of frames selected by CLIP, with associated relevance scores.
 - Frames showing detected table boundaries.
2. **Audio Analysis:**
 - Training & Validation Accuracy/Loss Curves for the audio CNN.
 - Confusion Matrix for audio event classification.
 - PCA of audio MFCC features, colored by class.
 - Examples of audio waveforms with detected event markers.



Different feature spaces because it is projecting two completely different feature spaces onto their first two principal components: the SVM plot projects 28 dim summary vectors (MFCC means/std, centroid, ZCR) that span large numeric ranges, whereas the CNN plots z-normalized MFCC spectrogram patches, yielding much smaller axis scales.



3. **Player Tracking and Detection:**

- Visualizations from player detection models showing detected players with confidence scores.
- Training curves (e.g., mAP, loss) for the custom YOLO player detector.
- Annotated video output showing players tracked with IDs and bounding boxes.
- Player position heat maps overlaid on a representation of the play area.

4. **Ball Tracking and Segmentation:**

- Training/validation loss and segmentation metric (e.g., Dice score) curves for the 3D U-Net.

Conclusion & Future Work

This paper has described the development of a multi-component system for table tennis analytics. The approach integrates CLIP-based data curation, foundation models for aspects of scene understanding like table and player detection, and specialized neural network architectures for dynamic tasks such as ball segmentation (3D U-Net) and audio event classification (2D CNNs). The system processes video and audio from match footage, prepares data for model training, and aims to analyze player movements and game events. Visualizations such as heatmaps and annotated videos are intended to aid in understanding game dynamics.

Future Work:

1. **System Integration and Event Synthesis:** Further work could focus on combining outputs from the individual modules to create a more unified timeline of game events.
2. **Exploration of Advanced Event Recognition:** Investigate methods to recognize more complex game events by potentially fusing audio cues, player actions, and ball trajectory information.
3. **Refinement of Scene Geometry:** Improve table and net detection to provide a more consistent frame of reference for spatial analysis.
4. **Stroke Classification Study:** Explore the possibility of classifying player strokes by analyzing player motion in conjunction with ball trajectory, potentially requiring player pose estimation.
5. **Optimization for Faster Processing:** Investigate model optimization and pipeline adjustments for nearer to real-time analysis.
6. **Evaluation of Robustness:** Assess and work on improving system performance across varied environmental conditions (lighting, camera angles).
7. **Benchmarking:** Where possible, benchmark components against standard datasets or establish clear metrics for internal evaluation.
8. **Development of an Interactive Interface:** Consider creating an interface for users to visualize and interact with the extracted analytics.
9. Integration of **weakly supervised approach** for more robust tracking along with more **aggressive data augmentation**
10. Engineer better algorithms for segmenting out the motion trajectory from noise trajectory using physics priors and clustering techniques derived from trajectory features.

Appendix

Model Weights & Dataset:

- Dataset:
<https://drive.google.com/drive/folders/1u9G2Tgl4oLG28WdlOwvWazqE9txYR9ha?usp=sharing>
- YOLOv11 for Player
detection: https://drive.google.com/drive/folders/1KhHpVIGQFXt2k0ohdm2vUiS7zHmyAQ_W?usp=sharing